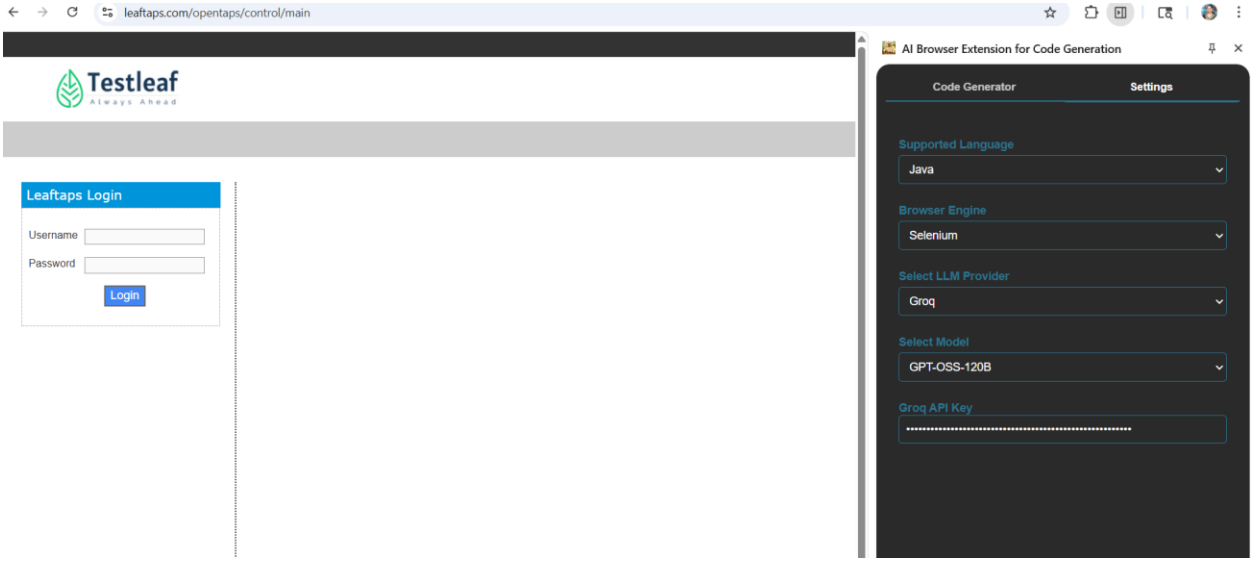
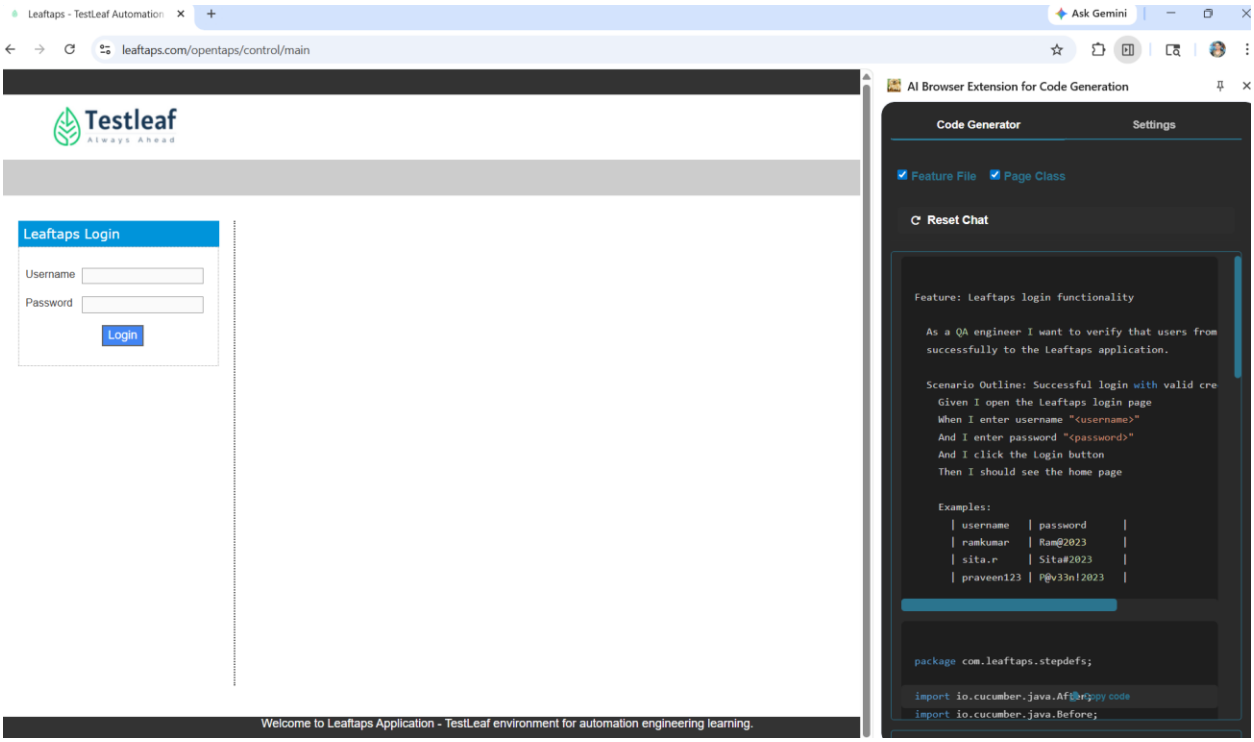
Selenium + Java and Feature File + Page Class





```
Feature: Leaftaps login functionality

 As a QA engineer I want to verify that users from South-India can log in
 successfully to the Leaftaps application.

 Scenario Outline: Successful login with valid credentials
  Given I open the Leaftaps login page
  When I enter username "<username>"
  And I enter password "<password>"
  And I click the Login button
  Then I should see the home page

  Examples:
   | username   | password     |
   | ramkumar   | Ram@2023     |
   | sita.r     | Sita#2023    |
   | praveen123 | P@v33n!2023  |
```

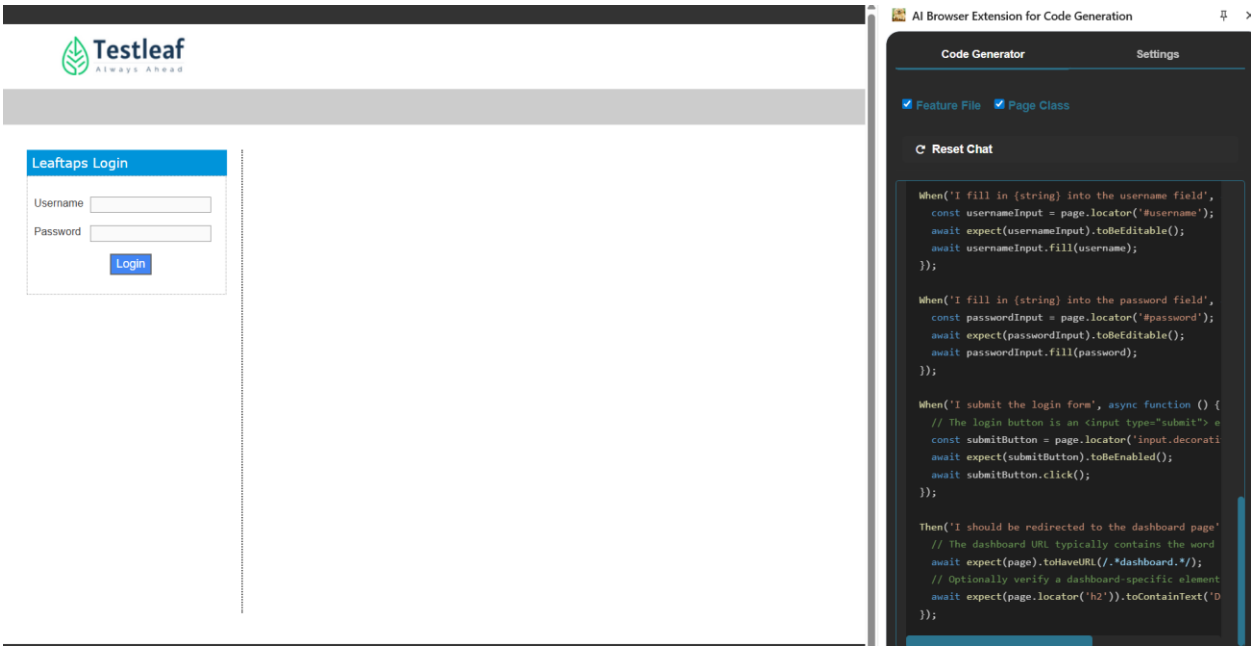
```
package com.leaftaps.stepdefs;

import io.cucumber.java.After;
import io.cucumber.java.Before;
import io.cucumber.java.en.*;
import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.*;
import java.time.Duration;
import static org.junit.Assert.*;
```

```
public class LoginStepDefinitions {

    private WebDriver driver;
    private WebDriverWait wait;

    @Before
    public void setUp() {
        // ChromeDriver executable must be available in the system PATH
        driver = new ChromeDriver();
        wait = new WebDriverWait(driver, Duration.ofSeconds(15));
        driver.manage().window().maximize();
    }

    @After
    public void tearDown() {
        if (driver != null) {
            driver.quit();
        }
    }

    @Given("I open the Leaftaps login page")
    public void openLoginPage() {
        driver.get("https://leaftaps.com/opentaps/control/main");
        // Ensure the login form is present before proceeding
        wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("login")));
    }
```

```
@When("I enter username {string}")
public void enterUsername(String username) {
   WebElement userField = wait.until(
         ExpectedConditions.elementToBeClickable(By.id("username")));
   userField.clear();
   userField.sendKeys(username);
}

@When("I enter password {string}")
public void enterPassword(String password) {
   WebElement passField = wait.until(
         ExpectedConditions.elementToBeClickable(By.id("password")));
   passField.clear();
   passField.sendKeys(password);
}

@When("I click the Login button")
public void clickLoginButton() {
   // The login button is an <input type="submit" value="Login">
   WebElement loginBtn = wait.until(
         ExpectedConditions.elementToBeClickable(
               By.xpath("//input[@type='submit' and @value='Login']")));
   loginBtn.click();
}

@Then("I should see the home page")
public void verifyHomePage() {
   // Typical post-login verification – presence of a "Logout" link
```

```
        By logoutLinkLocator = By.linkText("Logout");
        WebElement logoutLink = wait.until(
            ExpectedConditions.visibilityOfElementLocated(logoutLinkLocator));

        assertTrue("Logout link should be visible after a successful login",
            logoutLink.isDisplayed());
    }
}
```

Typescript+ Playwright and Feature File + Page Class:



```
Feature: Leaftaps login functionality

 As a QA engineer I want to verify that users from South India can log in
 successfully to the Leaftaps application.

 Scenario Outline: Successful login with valid credentials
  Given I open the Leaftaps login page
  When I fill in username "<username>"
  And I fill in password "<password>"
  And I submit the login form
  Then I should be redirected to the dashboard page

  Examples:
   | username     | password   |
   | krishnan.r   | K@12345    |
   | sita.rao     | S!t@2023   |
   | ravi.kumar   | Rav1#Pass  |
   | lakshmi.p    | Lakshmi$1  |
   | ashok.m      | Ash0k!2022 |
```

```
import { Before, After, Given, When, Then } from '@cucumber/cucumber';
import { chromium, Browser, BrowserContext, Page, expect } from '@playwright/test';

// ---------------------------------------------------------------------------
// Global Playwright objects – strict typing
// ---------------------------------------------------------------------------
let browser: Browser;
let context: BrowserContext;
```

```
let page: Page;

// ---------------------------------------------------------------------------
// Hooks – launch and close the browser once per scenario
// ---------------------------------------------------------------------------
Before(async function () {
  // Launch Chromium in headless mode; can be overridden via env vars if needed
  browser = await chromium.launch({ headless: true });
  context = await browser.newContext();
  page = await context.newPage();
});

After(async function () {
  // Clean up resources
  if (page) await page.close();
  if (context) await context.close();
  if (browser) await browser.close();
});

// ---------------------------------------------------------------------------
// Step definitions
// ---------------------------------------------------------------------------
Given('I open the Leaftaps login page', async function () {
  await page.goto('https://leaftaps.com/opentaps/control/main');
  // Ensure the login form is visible before proceeding
  await expect(page.locator('#login')).toBeVisible();
});
```

```
When('I fill in username {string}', async function (username: string) {
  const usernameInput = page.locator('#username');
  await expect(usernameInput).toBeEditable();
  await usernameInput.fill(username);
});

When('I fill in password {string}', async function (password: string) {
  const passwordInput = page.locator('#password');
  await expect(passwordInput).toBeEditable();
  await passwordInput.fill(password);
});

When('I submit the login form', async function () {
  // The submit button is an <input type="submit"> with class decorativeSubmit
  const submitBtn = page.locator('input.decorativeSubmit');
  await expect(submitBtn).toBeEnabled();
  await submitBtn.click();
});

Then('I should be redirected to the dashboard page', async function () {
  // The dashboard URL typically contains "/dashboard" – adjust the regex if needed
  await expect(page).toHaveURL(/.*dashboard.*/);
  // Optionally verify a dashboard-specific element
  const dashboardHeader = page.locator('h1', { hasText: 'Dashboard' });
  await expect(dashboardHeader).toBeVisible();
});
```